torch.linalg.cholesky_ex#

https://pytorch.org/docs/stable/generated/torch.linalg.cholesky_ex.html

Description:

Computes the Cholesky decomposition of a complex Hermitian or real symmetric positive-definite matrix. This function skips the (slow) error checking and error message construction of torch.linalg.cholesky(), instead directly returning the LAPACK error codes as part of a named tuple (L, info). This makes this function a faster way to check if a matrix is positive-definite, and it provides an opportunity to handle decomposition errors more gracefully or performantly than torch.linalg.cholesky() does. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. If A is not a Hermitian positive-definite matrix, or if it's a batch of matrices and one or more of them is not a Hermitian positive-definite matrix, then info stores a positive integer for the corresponding matrix. The positive integer indicates the order of the leading minor that is not positive-definite, and the decomposition could not be completed. info filled with zeros indicates that the decomposition was successful. If check_errors=True and info contains positive integers, then a RuntimeError is thrown.

Function Signature

```
torch.linalg.cholesky_ex(A, *, upper=False,
check_errors=False, out=None)#

Parameters

Keyword Arguments
```

Parameters

Keyword

upper (bool, optional) – whether to return an upper triangular matrix. The tensor returned with upper=True is the conjugate transpose of the tensor returned with upper=False. check_errors (bool, optional) – controls whether to check the content of infos. Default: False. out (tuple, optional) – tuple of two tensors to write the output to. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(2, 2, dtype=torch.complex128)
>>> A = A @ A.t().conj()  # creates a Hermitian positive-
definite matrix
>>> L, info = torch.linalg.cholesky_ex(A)
>>> A
tensor([[ 2.3792+0.0000j, -0.9023+0.9831j],
        [-0.9023-0.9831j,  0.8757+0.0000j]],
dtype=torch.complex128)
>>> L
tensor([[ 1.5425+0.0000j,  0.0000+0.0000j],
        [-0.5850-0.6374j,  0.3567+0.0000j]],
dtype=torch.complex128)
>>> info
tensor(0, dtype=torch.int32)
```

Notes & Warnings

NOTE Note When the inputs are on a CUDA device, this function synchronizes only when check_errors= True.

⚠ **WARNING** Warning This function is “experimental” and it may change in a future PyTorch release.

NOTE See also torch.linalg.cholesky() is a NumPy compatible variant that always checks for errors.

Detailed Documentation

Documentation

Computes the Cholesky decomposition of a complex Hermitian or real symmetric positive-definite matrix.

This function skips the (slow) error checking and error message construction of torch.linalg.cholesky(), instead directly returning the LAPACK error codes as part of a named tuple (L, info). This makes this function a faster way to check if a matrix is positive-definite, and it provides an opportunity to handle decomposition errors more gracefully or performantly than torch.linalg.cholesky() does.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

If A is not a Hermitian positive-definite matrix, or if it's a batch of matrices and one or more of them is not a Hermitian positive-definite matrix, then info stores a positive integer for the corresponding matrix. The positive integer indicates the order of the leading minor that is not positive-definite, and the decomposition could not be completed. info filled with zeros indicates that the decomposition was successful. If check_errors=True and info contains positive integers, then a RuntimeError is thrown.

When the inputs are on a CUDA device, this function synchronizes only when check_errors= True.

This function is "experimental" and it may change in a future PyTorch release.

torch.linalg.cholesky() is a NumPy compatible variant that always checks for errors.

A (Tensor) – the Hermitian n times n matrix or the batch of such matrices of size (*, n, n) where * is one or more batch dimensions.

upper (bool, optional) – whether to return an upper triangular matrix. The tensor returned with upper=True is the conjugate transpose of the tensor returned with upper=False.

check_errors (bool, optional) – controls whether to check the content of infos. Default: False.

out (tuple, optional) – tuple of two tensors to write the output to. Ignored if None. Default: None.